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Natural Compounds against Mpox: Mapping Evidence and Identifying Gaps

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ABSTRACT

The global spread of Mpox necessitates exploration of novel treatment options. Considering the established history of herbal medicine in managing infectious diseases, this study reviewed the literature on phytotherapy for Mpox, addressing gaps in evidence-based herbal interventions. A thorough search was conducted across the Scopus, PubMed, and Cochrane databases, as well as grey literature, up to August 2024 to retrieve studies on natural compounds with potential efficacy against Mpox and its associated symptoms. Data were analysed for publication characteristics, the compounds or herbal plants investigated, and their effects on the virus. A total of 37 articles with 242 citations were identified, demonstrating a steady increase in research activity since the first study in 2011, peaking in 2023 with 21 publications and 114 citations. The majority of studies originated from Southeast Asian countries. In terms of study design, most investigations were *in silico* (n = 31, 84%), followed by *in vitro* studies (n = 4, 11%), with no *in vivo* or clinical interventions reported. The primary focus was on the antiviral activities of natural products, with polyphenols identified as the most prevalent lead compounds. Whilst these findings highlight the growing interest in phytotherapy for Mpox, they also underscore the predominance of computational studies. To build upon this foundation of *in silico* evidence, further laboratory and animal studies are imperative for translating these insights into clinical applications. The comprehensive library of compounds gathered through this research provides a valuable resource to facilitate this crucial next step in the development of herbal interventions against Mpox.

Introduction

Monkeypox virus, an enveloped double-stranded DNA virus belonging to the Orthopoxvirus genus within the Poxviridae family, is the causative agent of Mpox. Initially discovered in monkeys in Denmark in 1958, the first recorded case in humans occurred in the Democratic Republic of the Congo in 1970. This neglected infectious disease culminated in a pandemic between 2022 and 2023, resulting in 85,473 cases and 89 mortalities in 110 countries [1]. Fifteen months after the conclusion of this emergency period, the World Health Organization declared Mpox a 'public health emergency of international concern' on 14 August 2024, in response to the circulation of a newer and more virulent strain designated as clade 1b [2]. This new declaration came amidst concerning trends in several regions, including some endemic areas. For instance, the Democratic Republic of the Congo reported a high number of cases, including an annual incidence of 1146 per 100,000 in 2023 and continued high case numbers in 2024 [3, 4].

The transmission of Mpox was suggested to be from zoonotic sources and contact with infected animals, but it is postulated that the new clade is spreading via sexual contact [5]. This disease may lead to dermatologic, inflammatory, respiratory, ophthalmic, respiratory, or gastrointestinal complications, especially in patients with weak immune systems, a condition that may be prevalent in affected populations [6, 7]. In settings with limited resources where vaccines are unavailable, early intervention and supportive care are vital for alleviating symptoms like pain and

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rash and preventing complications in Mpox patients. The public health risk of resistant strains, such as clade 1b, underscores the need for ongoing treatment exploration [8]. Phytotherapy and natural products, with their long history of managing symptoms and demonstrating anti-infective efficacy, could be beneficial in these challenging situations [9].

A preliminary search of the PubMed database retrieved no studies specifically addressing natural compounds for the management of Mpox (see supplementary material, preliminary search). Therefore, this novel study aimed to systematically investigate the literature for natural compounds with potential efficacy against Mpox or its associated symptoms.

Results

A total of 37 articles, encompassing 242 citations, were retrieved (see supplementary material, **Figure 1S** and **Table 1S**). The first study was published in 2011. Since then, the net number of publications and citations has steadily increased, with 2023 recording the most citations (n = 114) and publications (n = 21) (see supplementary material, **Figure 2S**).

According to the World Health Organization's regional classifications, researchers from the South-East Asian and Eastern Mediterranean Regions contributed the majority of studies in this field, with Saudi Arabia, India, and China leading in terms of international collaborations. Notably, nearly all institutions had a single publication, indicating diverse contributions from various entities. Whilst several studies reported no funding, others received support from a range of sources, particularly from the authors' institutions.

All studies were at the preclinical stage, with *in silico* investigations comprising the majority. Additionally, there were four *in vitro* studies, with only one conducted after the 2022 Mpox pandemic. The literature revealed no *in vivo* or clinical interventions regarding phytotherapy or the utilisation of natural compounds in the management of Mpox.

The predominant focus of the literature was on phytotherapy. Among the lead phytochemicals identified, polyphenols, encompassing both aglycones and glycosides, were the most prevalent, with quercetin, rutin, and kaempferol cited as the most frequently mentioned compounds. Other notable classes included alkaloids, terpenoids, and tannins, each represented by at least one compound with a solitary occurrence. Additionally, natural bacteriocins, specifically glycocin F and lactococcin G, were reported. In terms of the mechanisms studied, most compounds exhibited effects on profilin-like protein, particularly the A42R protein, followed by thymidine kinase and DNA-dependent RNA polymerase of the monkeypox virus. Further details can be found in the supplementary material, **Table 2S**.

From an initial set of 823 author-provided and index keywords, a refined subset of 26 keywords was identified, each with a minimum occurrence of 5 in the literature. By removing redundant keywords, a graph was generated that illuminates the dominance of molecular docking and its correlation with keywords such as monkeypox, natural product, antiviral agent, and unclassified drugs. This graph further emphasised the preclinical focus of the literature pertaining to natural compounds as a potential treatment for Mpox (see supplementary file, **Figure 3S**).

Discussion

This study constitutes the first comprehensive examination of the literature regarding natural compounds with potential efficacy against Mpox via a systematic approach. The findings indicate a predominant emphasis on molecular docking studies of phytochemicals, particularly flavonoids and alkaloids, alongside reports of the potential efficacy of microbial-derived natural compounds against Mpox. However, there remains a notable scarcity of *in vitro* investigations and a complete absence of *in vivo* or clinical studies within the current research landscape.

Translational research is necessary to introduce findings from basic science to practical applications in the clinical practice [10]. However, the literature appears to have reached a saturation point regarding *in silico* investigations aimed at identifying natural lead compounds effective against Mpox, underscoring the necessity for further evaluation of these compounds through laboratory and animal studies as the next logical step. Whilst the biohazard nature of Mpox presents significant challenges for laboratory research, future investigations should prioritise study designs that could facilitate critical translation points in clinical medicine. Such endeavours are essential for advancing promising compounds from the laboratory bench to the bedside, enhancing patient care and therapeutic outcomes. Otherwise, the research landscape risks becoming inundated with numerous natural compounds, with only a select few undergoing practical investigations against this global threat.

Materials and Methods

A comprehensive search was conducted across PubMed, Scopus, and the Cochrane Database of Systematic Reviews from inception until 26 August 2024, utilising an exhaustive list of keywords related to 'Mpox', 'herbal medicine', and 'natural compounds'. Additionally, preprints were obtained through the Scopus database. No filters were applied, allowing studies in all languages to be considered eligible for inclusion. Studies lacking original data, such as narrative reviews with no clear methodology, those focused on vaccine development, or those addressing epidemiological and surveillance systems were excluded from the analysis. Furthermore, ClinicalTrials.gov and the World Health Organization International Clinical Trials Registry Platform were also searched for relevant studies. All identified studies and literature from these databases were collated into a single file using Microsoft Excel Version 2108. The data were analysed for publication and citation trends, compounds or herbal plants investigated, the effects of these compounds on the Mpox virus, associated keywords, and basic characteristics, including author names, publication years, funding sources, and study designs. Data visualisation techniques were performed using Microsoft Excel and VOSviewer software version 1.6.20 (Leiden University, Leiden, Netherlands).

Supporting Information

The search strategy, reasons for excluding articles on the full-text level, publication and citation trends, a detailed list of eligible studies and natural compounds, and the association of keywords in the literature are available as supporting information.

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Contributors' Statement

A. Sharifan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Supervision, Project administration.

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Conflict of Interest

AS holds an unpaid membership role in the Cochrane Early Career Professionals Steering Group and is a spoke leader of the Cochrane Planetary Health Thematic Group.

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